Module 4 Assignment: Structure-Based Druggability of PDACTargets

CEL (P19835) and XDH (P47989) were selected from the Module 2 PDAC ranking for structural assessment. CEL ranked fourth in Module 2 and first after Module 3 reranking, with a small-molecule prior of 0.716; it is a compact serine hydrolase whose predicted catalytic pocket was the open question. XDH ranked ninth and sixth respectively, with a prior of 0.503, and was chosen as a positive control because its active site is validated by two approved drugs. This analysis tests whether their predicted structures contain credible pockets; it does not establish whether inhibiting either protein would benefit PDAC.

1. Which structural regions can be trusted?

Target	Mean pLDDT	Above 90	Below 50	Stretches below 50 (10+ residues)
CEL	82.6	66.3%	16.2%	3-17; 555-592; 717-753
XDH	96.0	93.4%	1.7%	168-190

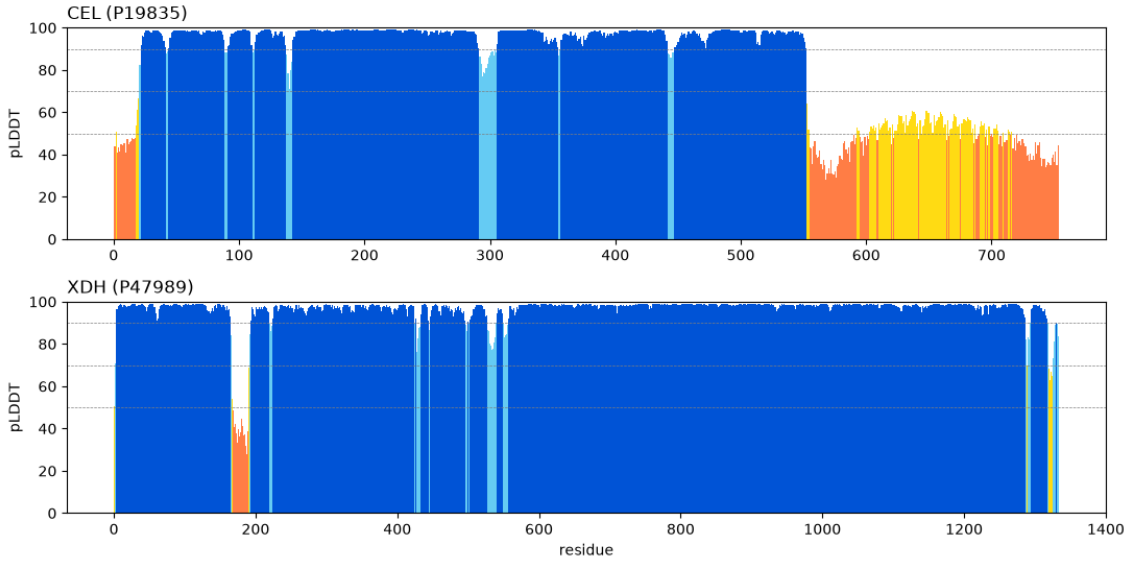


Figure 1. pLDDT profiles. CEL has a confident central region and a less reliable C-terminal region. XDH is predominantly high confidence.

For CEL, the PAE-based finder grouped residues 21-552 into one confident region containing 532 residues. Applying the decision table, this interval is the region that can be trusted for pocket analysis, regions with low confidence ends are not interpretable. CEL is secreted and has a glycosylated, repeat rich C-terminal region [3]. Its low confidence tail is therefore consistent with a region whose biological context is poorly represented by a single unglycosylated model. The N-terminus overlaps the signal-peptide region described for CEL [3].

For XDH, the finder returned one confidence group spanning 2-1333, containing 1300 confident residues. By the same decision table, most of the model is suitable for local pocket analysis, but residues 168-190 should be treated cautiously. This interval corresponds to the loop whose proteolysis or disulphide rearrangement converts the dehydrogenase form of the enzyme into the superoxide-generating form, so it is a region with a documented conformational switch rather than a modelling artefact. A single PAE-connected group does not mean XDH has only one biological domain. For both targets, inter-domain PAE was unavailable because the algorithm found no second group.

2. **Agreement with experimental structures**

Residues were paired by sequence alignment, and identical C-alpha pairs with AlphaFold pLDDT of at least 70 were superimposed. CEL was compared with 1F6W chain A and XDH with 2E1Q chain A. These comparisons validate the aligned regions rather than the complete full-length proteins.

Target / reference	Confident identical pairs	RMSD	Largest reported deviations
CEL / 1F6W A	527	1.35 Å	444-449; maximum 9.1 Å at residue 447
XDH / 2E1Q A	1,297	0.48 Å	3 (5.6 Å), 537 (3.6 Å), 538 (3.4 Å), 1288 (2.7 Å)

CEL shows good overall agreement but a substantial local difference around residues 444-449. This can be interpreted as a conformational difference rather than a prediction error. The deviation profile is below 1 Å almost everywhere, including the β-sandwich core and the catalytic triad, and the excursion is confined to a six-residue surface loop remote from the active site. Terzyan and colleagues report that several surface loops around the active site of 1F6W are mobile and poorly ordered [1], and 1F6W carries two engineered surface substitutions introduced to assist crystallization, so crystal-contact effects on surface loops are expected. CEL's activity is also regulated by a surface loop that partly shields the active site until bile salt binds; no bile salt is present in either structure, so neither represents the activated state. 1F6W is a catalytic-domain structure with no small-molecule ligand for this check [1], and its agreement cannot validate CEL's missing C-terminal tail.

XDH shows very close agreement over most aligned residues. However, 2E1Q contains the E803 active site mutation and bound cofactors and SAL [2], whereas the AlphaFold model represents a single protein chain without these bound components. The differences here are therefore mostly things the crystal has that the prediction lacks, not errors in the prediction. Excluding the mismatched residue from fitting does not remove possible neighboring effects. The small RMSD supports the fold but does not establish identical active-site chemistry.

3. **Pocket confidence and druggability calls**

Pockets with mean lining-residue pLDDT below 70 were rejected. The table shows the top five pockets per target ranked by druggability score. pockets with mean internal PAE above 10 Å were marked suspect, and the remainder were kept.

Target	Rank	Pocket	Score	Mean pLDDT	PAE (Å)	Verdict
CEL	1	56	0.925	93.7	2.1	keep
CEL	2	60	0.562	51.1	13.4	reject
CEL	3	3	0.193	98.5	1.2	keep
CEL	4	62	0.135	95.6	2.2	keep
CEL	5	11	0.093	98.2	1.2	keep
XDH	1	2	0.979	98.2	1.5	keep
XDH	2	1	0.933	96.6	2.1	keep
XDH	3	85	0.176	98.2	1.4	keep
XDH	4	3	0.074	96.1	1.8	keep
XDH	5	84	0.060	97.3	1.6	keep

CEL: Pocket 56 has a strong score and passes both confidence checks, and the drop to the next pocket is steep (0.193), so this is a single well-defined site rather than a cloud of marginal cavities. Pocket 60 is rejected despite its second-place score, showing why geometry alone is insufficient, its lining residues average pLDDT 51.1 and PAE 13.4 Å, placing it in the low-confidence C-terminal region rather than the trusted 21-552 interval.

XDH: Pocket 2 combines a score of 0.979, high lining-residue confidence, low PAE, and overlap with experimental ligand contact residues. This supports structural tractability while leaving affinity, selectivity, and therapeutic benefit unresolved.

4. **Experimental ligand-site check**

CEL's selected reference, 1F2W chain A, had no eligible bound small molecule, so its pocket could not be checked against an experimental ligand site. For XDH, the 4.5 Å contact cutoff and sequence-based numbering map were used, and every detected pocket was searched.

Ligand near XDH chain A	Mapped site residues	Coverage by best kept pocket 2	Best matching pocket / rank	Site coverage
DYI A7003	27	74%	2 / 1	74%
FAD A2003	39	0%	1 / 2	90%
SAL A2006	12	92%	2 / 1	92%

The best kept pocket is therefore at the SAL- associated site and overlaps the DY1 cofactor environment. The FAD site is distinct, the pocket 1 covers 90% of its mapped contacts, although pocket 2 covers none. DY1 and FAD are cofactors, so their sites should not be described as interchangeable drug-binding sites [2]. SAL A2006 is the relevant SAL instance

contacting the selected protein chain A in the saved output. Other copies associated with other protein chains should not be pooled into this chain-specific comparison.

Coverage measures shared contact residues, not ligand pose recovery or binding strength. SAL contact residue 803 also lies at the engineered E803V position, adding uncertainty to transfer of this site to the wild-type protein. A useful next step is to inspect pocket 2, the SAL contacts, and residue 803 together and compare a wild-type experimental reference.

No severe scoring failure was apparent for these XDH sites, their best-matching pockets ranked first and second.

5. Target-specific limits and how to address them

The model lack important biological features, including CEL glycosylation and XDH cofactors and protein partners. These omissions may affect pocket accessibility. Comparing more complete experimental structures and testing binding and enzyme activity would help address this limitation.

Only one predicted conformation was assessed, although protein movement may change pocket shape or accessibility. Comparing several structures or using molecular dynamics could help identify stable pockets.

CEL was validated only against a catalytic-domain construct, while the XDH reference contains an active-site mutation. High pLDDT also does not confirm drug binding or function.

6. Integration with Modules 2 and 3

The structural results support the earlier small-molecule predictions, but XDH showed stronger experimental support despite CEL's higher Module 3 prior. CEL moved from rank 4 to rank 1, with a small-molecule prior of 0.716, while XDH moved from rank 9 to 6, with a prior of 0.503.

Both targets were retained for follow-up, with XDH prioritized for binding studies. CEL requires further testing to confirm whether pocket 56 has a functional role. Weaker pocket validation alone does not justify switching to an antibody approach.

Both targets were downregulated in Module 2, so finding a pocket does not establish that inhibition would benefit PDAC. Reduced CEL expression may also reflect changes in tissue composition rather than tumor-cell biology. Experiments in PDAC and normal pancreatic models would help determine whether either target should be inhibited or restored.

7. **Anchor check: p53**

The AlphaFold p53 model matched 1TUP chain B with an RMSD of 0.68 Å over 192 confident residues (pLDDT >= 70), supporting the aligned DNA-binding core. C176, H179, C238, and C242 coordinated zinc at 2.1-2.3 Å, while R248 and R273 contacted DNA at 3.4 Å and 2.5 Å. R249 was nearby at 7.2 Å, whereas R175, G245, Y220, and R282 met neither contact cutoff. These findings help explain how mutations can disrupt zinc dependent stability or DNA binding despite high confidence in the wild-type structure. R175H's reported 19th-percentile Module 3 score also illustrates that sequence-based scores may not fully capture indirect destabilization near the zinc site. Structural context therefore helps interpret variant scores, but neither approach alone establishes each mutation's exact effect.

REFERENCE

1] Terzyan, S., Wang, C. S., Downs, D., Hunter, B., & Zhang, X. C. (2000). Crystal structure of the catalytic domain of human bile salt activated lipase. Protein Science, 9, 1783-1790. https://doi.org/10.1110/ps.9.9.1783. Structure: https://www.rcsb.org/structure/1F6W

[2] Yamaguchi, Y., Matsumura, T., Ichida, K., Okamoto, K., & Nishino, T. (2007). Human xanthine oxidase changes its substrate specificity to aldehyde oxidase type upon mutation of amino acid residues in the active site: Roles of active site residues in binding and activation of purine substrate. Journal of Biochemistry, 141, 513-524. https://doi.org/10.1093/jb/mvm053. Structure: https://www.rcsb.org/structure/2E1Q

[3] Landberg, E., et al. The mucinous domain of pancreatic carboxyl-ester lipase (CEL) contains core 1/core 2 O-glycans that can be modified by ABO blood group determinants. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC6302156/

AI Use Disclosure

Claude (Anthropic, 2026) was used to structure the report and to mitigate errors beyond those noted in the assignment module. All content and analysis are my own. Anthropic. (2026). Claude (Opus 4.5) [Large language model]. https://claude.ai